AYE

• MECHANICAL ENGINEERING • LESSON 07

Machine Elements

Shafts • Bearings • Gears • Belts • Fasteners

Selection | Design | Standards | Applications

What You Will Learn

- Shaft design: materials, standard diameters, keyways and stress analysis
- Bearings: rolling element vs journal, selection by load and speed (C/P method)
- Gear types: spur, helical, bevel, worm — selection and gear ratio calculation
- Belt, chain, and spring selection — plus fastener grades and torque values

Gear Ratio Formula

$$VR = N_{driven} \div N_{driver}$$

$$VR = T_{driver} \div T_{driven} \quad \text{(teeth ratio)}$$

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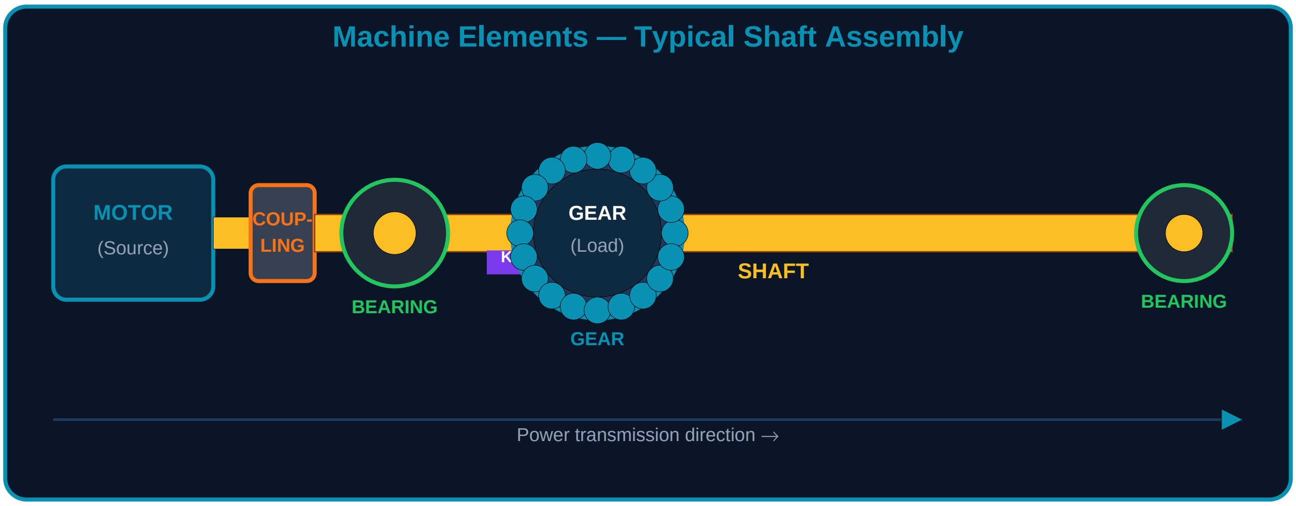
MECH-007: Machine Elements

Machine elements are the standard components from which all mechanical systems are assembled. Every pump, gearbox, conveyor, lathe, vehicle drivetrain, and industrial machine is built from a finite set of these elements. Understanding how to select, size, and apply them correctly is foundational knowledge for every mechanical engineer.

Learning Objectives

- Describe the function of shafts, keys, couplings, bearings, gears, belts, springs, and fasteners
- Select appropriate shaft diameter based on torsional shear stress and standard sizes
- Choose bearing type and size using the basic dynamic load rating C and calculated load P
- Calculate gear velocity ratio and select gear type for given shaft arrangement and reduction requirement
- Apply correct fastener grade and calculate tightening torque for critical bolted joints



Machine Elements — Typical Shaft Assembly

1. Shafts — The Backbone of Power Transmission

A shaft is a rotating member that transmits torque from a power source (motor, engine) to driven components (gears, pulleys, impellers). Shafts are subjected to torsion, bending, and sometimes axial loads simultaneously. Design involves selecting both material and diameter to keep stresses within safe limits.

Shaft Parameter	Formula / Standard	Notes
Torsional shear stress	$\tau = 16T / (\pi \times d^3)$ in Pa	T = torque (N·m); d = shaft diameter (m). τ must be ≤ allowable shear stress of material.
Power-torque relationship	$T = P \times 60 / (2\pi \times N)$	P = power (W); N = speed (rpm). Convert to SI before using.
Minimum diameter (torsion only)	$d \geq$ ■$(16T / (\pi \times \tau_{allow}))$	For steel shafting: $\tau_{allow} \approx$ 40–60 MPa for rotating shafts with keyways.

Standard shaft diameters (ISO)	Preferred: 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70 mm	Always round UP to next standard size after calculation.
Common shaft materials	Medium carbon steel: C40, C45 (most common). Alloy: 40Cr, 42CrMo for high stress.	C45: $\sigma_y \approx 370$ MPa, $\tau_{allow} \approx 45$ MPa. Use alloy steel for keyways and splines.

Keys and Keyways

A key is a machine element that transmits torque between a shaft and a hub (gear, pulley, sprocket). The key fits into matching keyways cut into both the shaft and hub. The most common type is the **parallel (square or rectangular) key** — selected from ISO 773/774 tables based on shaft diameter.

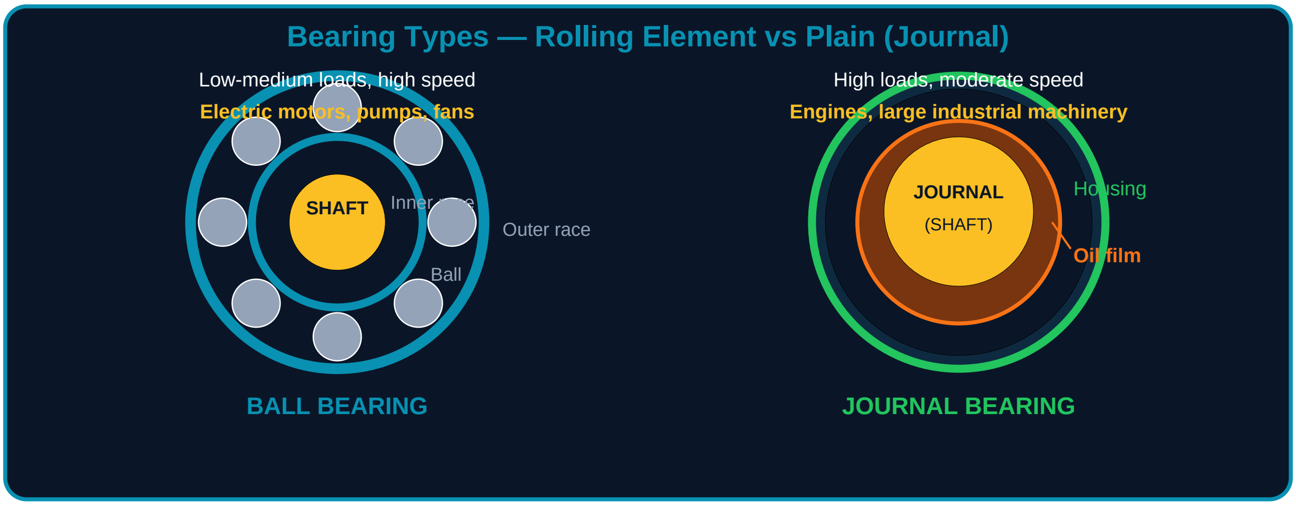
Shaft Diameter (mm)	Key Width × Height (b×h mm)	Keyway Depth in Shaft	Application
17 — 22 mm	6 × 6	3.5 mm	Small motors, fans, light duty
22 — 30 mm	8 × 7	4.0 mm	Pumps, gearbox input shafts
30 — 38 mm	10 × 8	5.0 mm	Medium gearboxes, conveyors
38 — 44 mm	12 × 8	5.0 mm	Industrial motors, blowers
44 — 50 mm	14 × 9	5.5 mm	Large pumps, heavy conveyors
50 — 58 mm	16 × 10	6.0 mm	Main drive shafts, agitators

2. Couplings — Connecting Shafts

A coupling connects two shafts to transmit torque. The type chosen depends on whether the shafts are perfectly aligned, whether misalignment must be accommodated, and whether vibration isolation or torque limiting is required.

Coupling Type	Misalignment Tolerance	Best Application
Rigid (sleeve or flange)	Zero — shafts must be perfectly aligned	Precisely aligned shafts where no relative movement occurs (vertical pump shafts, test rigs)
Jaw / Spider (flexible)	Angular ±1°, parallel 0.5mm, axial ±1mm	General industrial drives — electric motors to gearboxes, pumps, fans. Elastomeric spider absorbs vibration.
Disc / Bellows coupling	Angular ±0.5°, low parallel, good axial	Servo drives, encoders, precision applications. No backlash. Transmits torque without lubrication.
Universal joint (Cardan)	Angular up to ±45° depending on type	Vehicle drivetrains, steering columns, industrial drives with large angle between shafts.
Fluid coupling	No mechanical connection — hydraulic	Soft-start drives (conveyors, crushers) to protect motor from start-up shock load. Overload protection.

3. Bearings — Supporting Rotating Shafts



Bearing Type	Radial Load	Axial Load	Speed	Lubrication	Typical Application

Deep groove ball (6000 series)	Medium	Light-medium both directions	Very high	Grease or oil	Electric motors, pumps, gearboxes. Most common type.
Angular contact ball (7000)	Medium	High — one direction	High	Oil preferred	Machine tool spindles, ball screws. Usually in pairs.
Cylindrical roller (NU/NJ)	High	None (NU) or limited (NJ)	High	Oil preferred	Gearboxes, rolling mills, heavy duty radial loads.
Tapered roller (30000)	High	High — one direction	Medium	Oil	Vehicle wheel hubs, bevel gearboxes. Always in pairs.
Spherical roller (22000)	Very high	Both directions, self-aligning	Medium	Oil or grease	Conveyors, crushers, paper mills — where shaft misalignment is unavoidable.
Plain (journal) bearing	Very high	Limited	Low-medium	Continuous oil feed	Crankshafts, large slow rotating shafts. Low cost for very heavy loads.

Bearing Selection — The C/P Method

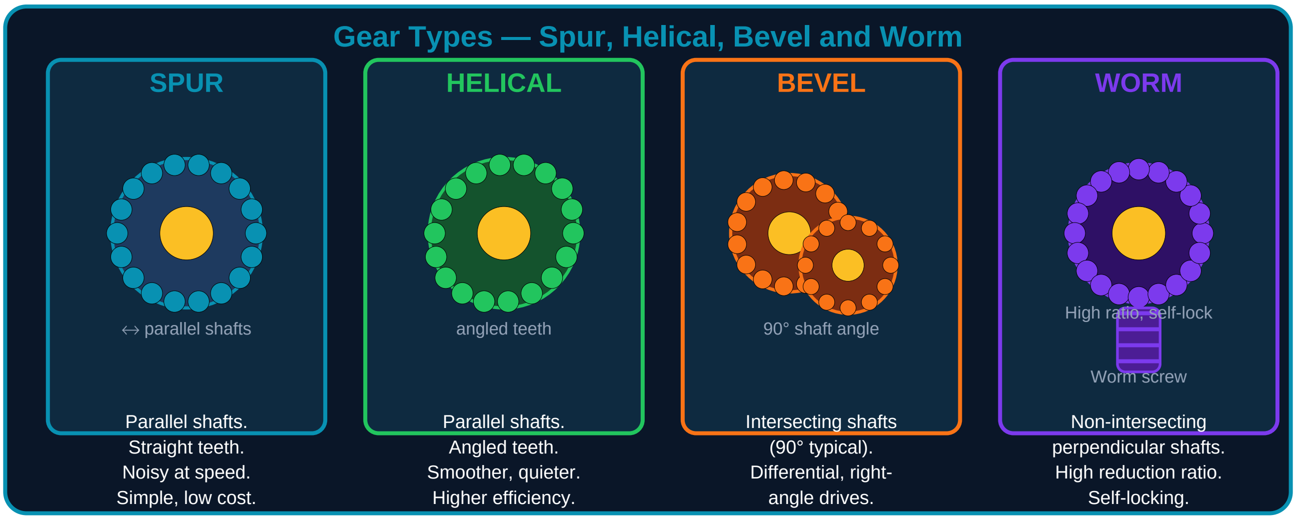
Basic Dynamic Load Rating C (from bearing catalogue) is the load that gives a basic rating life of one million revolutions. The required C is calculated from the equivalent dynamic load P and the desired life L■■:

C_required = P × (L■■h × 60 × N / 10■) ^ (1/p)

Where: P = equivalent radial load (N); L■■h = required life in hours; N = speed (rpm); p = 3 for ball bearings, 10/3 for roller bearings. Select a bearing where C_catalogue ≥ C_required.

4. Gears — Changing Speed and Torque

Gears are toothed wheels that mesh together to transmit motion and force between shafts. The fundamental relationship: when two gears mesh, larger gear = lower speed, higher torque. Smaller gear = higher speed, lower torque. Power is conserved (minus friction losses).



Gear Ratio Calculations

Velocity Ratio (VR) = N■ / N■ = T■ / T■ = d■ / d■

Where N = speed (rpm), T = number of teeth, d = pitch diameter. Output torque: $T_{out} = T_{in} \times VR \times \eta_{gear}$ ($\eta \approx 0.97$–0.99 for helical gears).

Gear Type	Shaft Arrangement	Typical VR Range	Efficiency	Key Advantage
Spur gear	Parallel, same plane	1:1 to 6:1 per stage	98–99%	Simplest, lowest cost, easy to manufacture. Noisy at high pitch-line velocity.
Helical gear	Parallel, same plane	1:1 to 10:1 per stage	97–99%	Quieter and smoother than spur. Axial thrust force requires thrust bearing.
Double helical (herringbone)	Parallel, same plane	1:1 to 10:1 per stage	98–99%	Cancels axial thrust from helical gear. Used in heavy gearboxes and turbine drives.
Bevel gear (straight)	Intersecting, usually 90°	1:1 to 5:1	97–99%	Right-angle drive. Used in differentials, hand drills, marine gearboxes.
Spiral bevel	Intersecting, usually 90°	1:1 to 5:1	97–99%	Smoother and quieter than straight bevel. Standard for automotive axles.
Worm gear	Non-intersecting, 90°	5:1 to 100:1 per stage	50–90%	Very high reduction ratio in single stage. Self-locking in most designs. Low efficiency at high ratios.

5. Belt and Chain Drives

Belt and chain drives transmit power between shafts that may be separated by large distances — where a gearbox would be impractical or too expensive. They also provide vibration damping (belts) or positive engagement without slip (chains).

Drive Type	Slip?	Speed Ratio	Max Power	Maintenance	Best Application
V-belt (wedge belt)	Yes — slight slip	Up to 7:1 single belt	Up to ~600 kW (multiple belts)	Tension check every 3–6 months. Replace at 15,000 hrs.	HVAC fans, compressors, agricultural equipment, general industry.
Timing belt (toothed)	No—positive drive	Up to 12:1	Medium power (<100 kW)	Low maintenance. Inspect for wear/cracks. Replace at 60,000–100,000 km (automotive).	Camshafts, precision conveyors, servo drives, office equipment.
Roller chain	No—positive drive	Up to 7:1	Very high (>1,000 kW)	Lubrication critical. Re-tension at 2% elongation. Clean and re-lubricate regularly.	Motorcycles, bicycles, conveyors, agricultural machinery, hoists.
Silent chain (inverted tooth)	No—positive drive	Up to 6:1	High	Requires lubrication, enclosed housing preferred.	Quiet high-speed chain drives — engines, machine tools.

6. Springs — Storing and Releasing Energy

Spring Type	Primary Function	Key Parameter	Typical Application
Helical compression spring	Resist compression force. Returns to free length when load removed.	Spring rate $k = F/\delta$ (N/mm). Solid length must not be exceeded.	Valve springs, safety valves, dies, mattresses, vehicle suspension
Helical tension spring	Resist stretching force. Stores energy when extended.	Initial tension (pre-load) + k. Hook failure is the most common failure mode.	Garage doors, safety interlocks, return mechanisms

Torsion spring	Resist angular deflection — applies a torque.	Spring rate in N·m/degree or N·m/radian.	Clothes pegs, mouse traps, vehicle suspension torsion bars, hinges
Belleville washer (disc spring)	High force in very small deflection. Stiffness adjustable by stacking.	Load-deflection characteristic is highly nonlinear.	Bolt pre-load maintenance, safety valves, heavy machinery
Leaf spring	Absorb shock and support load with large deflection capability.	Multiple leaves (blades) of different lengths for progressive rate.	Heavy vehicle suspension, railway rolling stock

7. Fasteners — Bolts, Nuts and Screws

Fasteners are among the most critical machine elements — incorrectly specified or torqued fasteners are a leading cause of joint failure in mechanical and pressure equipment. Key parameters are property class (strength grade), thread form, and tightening torque.

Property Class	Min. Tensile Strength (MPa)	Min. Yield Strength (MPa)	Material	Typical Application
4.6	400	240	Low carbon steel	Non-critical, light-duty structural joints
8.8	800	640	Medium carbon steel, quenched and tempered	General engineering — most common class. Machine frames, gearboxes.
10.9	1040	940	Alloy steel, Q&T;	High-strength joints — engine cylinder heads, critical structural.
12.9	1220	1100	Alloy steel, Q&T;, highest grade	Critical fastening — aerospace, racing engines, precision machinery.
A2-70 (stainless)	700	450	Austenitic stainless 304	Food industry, marine, chemical — where corrosion resistance is needed.

Bolt Tightening Torque

$$T_{tighten} = K \times d \times F_i$$

Where: K = torque coefficient (0.2 for dry steel, 0.15 for lubricated); d = bolt diameter (m); F_i = desired preload (N). Recommended preload = 70–80% of proof load. **Never reuse torque-to-yield (TTY) bolts.**

8. Practice Exercises

#	Exercise	Data Given
1	Shaft design for torsion	A shaft transmits 15 kW at 1450 rpm. Allowable shear stress = 45 MPa. Calculate: (a) torque T in N·m, (b) minimum shaft diameter, (c) standard ISO diameter to select.
2	Bearing selection	A deep groove ball bearing must support a radial load of 4,500 N at 1450 rpm for 20,000 hours. Calculate required C rating. Select from 6206 (C=19,500N), 6306 (C=28,100N), 6406 (C=47,500N).
3	Gear ratio calculation	A spur gear drive: driver gear 18 teeth at 1450 rpm, driven gear 72 teeth. Calculate: (a) velocity ratio, (b) output speed, (c) output torque if input torque = 50 N·m (η=0.98).

4	Belt drive design	A V-belt drive connects a 1450 rpm motor to a fan at 580 rpm. Motor pulley diameter = 150mm. Calculate required fan pulley diameter. If centre distance = 600mm, estimate belt length using: $L \approx 2C + \pi(D\blacksquare+D\blacksquare)/2 + (D\blacksquare-D\blacksquare)^2/4C$.
5	Fastener torque	An M16 bolt (Property Class 8.8) clamps a flange. Proof load = 72.9 kN (from table). Target preload = 75% of proof load. K = 0.20 (dry). Calculate required tightening torque in N·m.

Lesson Summary

Machine Element	Key Selection Rule
Shaft	Calculate τ from power and rpm. Select standard ISO diameter above calculated minimum. Material: C45 for most shafts; alloy steel for high-stress or keyways.
Key	Select from ISO 773 table by shaft diameter. Verify shear stress on key $\leq$ allowable for key material.
Coupling	Rigid: perfectly aligned shafts. Flexible jaw: standard motor-to-gearbox. Universal joint: angular offsets. Fluid: soft-start requirement.
Bearing	Ball bearing: high speed, medium load. Roller: high radial load. Tapered: combined radial+axial. Calculate $C_{req} = P \times (L\blacksquare\blacksquare \times 60N/10\blacksquare)^{(1/p)}$. Select $C_{cat} \geq C_{req}$.
Gear	Spur/helical: parallel shafts. Bevel: 90° intersecting. Worm: high ratio, perpendicular. $VR = T\blacksquare/T\blacksquare$. Output torque = input × VR × η.
Belt/Chain	V-belt: vibration damping, slight slip acceptable. Timing belt: no slip, moderate power. Roller chain: high power, positive drive. All require tension check.
Spring	Compression: resists compression. Tension: resists extension. Torsion: resists rotation. Key parameter: spring rate $k = F/\delta$.
Fastener	Grade 8.8 for general engineering. Grade 10.9/12.9 for critical joints. Torque = $K \times d \times F_{preload}$. K=0.20 dry, 0.15 lubricated.

What Is Next — MECH-008

MECH-008: Manufacturing Processes — Primary shaping (casting, forging, rolling, extrusion), machining operations (turning, milling, drilling, grinding), joining processes (welding, brazing, adhesive bonding), and surface finishing. Process selection based on material, shape, quantity, and tolerance requirements.

Resource	Details
MECH PDF Series	MECH-001 through MECH-030 — ayetechub.com/pdfs.html
Engineering Formulas	Free reference card — ayetechub.com/downloads/engineering-formulas.html

Community	Telegram: t.me/ayetechub